

Employee Attrition Prediction and Risk Scoring Project

1. Project Statement

Palo Alto Networks currently lacks a systematic, data-driven way to anticipate employee attrition. HR teams are often caught off guard by sudden resignations, including from high-performing or critical employees, and retention efforts tend to be reactive, such as counter-offers made after an employee has already decided to leave. There is no quantitative risk score to flag at-risk employees in advance, and no clear visibility into the key factors driving attrition. This project builds a machine learning model to predict the likelihood of an employee leaving the organization and generate risk scores that enable early identification of high-risk employees, targeted retention interventions, and data-driven workforce planning.

2. Data Description

Size: The dataset contains 1,470 rows and 31 columns

Attribute Groups:

- **Demographics:** Age, Gender, MaritalStatus, DistanceFromHome
- **Education:** Education, EducationField
- **Job Profile:** Department, JobRole, JobLevel, BusinessTravel
- **Compensation:** DailyRate, HourlyRate, MonthlyIncome, MonthlyRate, PercentSalaryHike, StockOptionLevel
- **Satisfaction & Engagement:** EnvironmentSatisfaction, JobSatisfaction, RelationshipSatisfaction, JobInvolvement, WorkLifeBalance
- **Tenure & Career Progression:** TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, NumCompaniesWorked, TrainingTimesLastYear
- **Performance & Work Conditions:** PerformanceRating, OverTime
- **Target Variable:** Attrition

Tools & Libraries:

- Python (pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, catboost, scipy, statsmodels)
- Jupyter Notebook / VS Code
- Streamlit

3. Methodology

3.1 Data Overview & Sanity Checks

The dataset was checked for missing values and duplicate rows; none were found. Since the dataset had no unique identifier, a surrogate primary key (EmplId) was created.

3.2 Exploratory Data Analysis (EDA)

To avoid repetitive, column-by-column analysis across 31 attributes, EDA was conducted using the thematic attribute groups defined in the Data Description (Demographics, Education, Job Profile, Compensation, Satisfaction & Engagement, Tenure & Career Progression, Performance & Work Conditions). Within each group, distributions were visualized together using grouped grid plots, histograms for numeric variables and count plots for categorical variables, to identify skew, spread, and class balance within features.

3.3 Feature Engineering & Selection

Statistical tests were used to determine which attributes had a significant relationship with Attrition, and only significant attributes were retained for modeling. Numeric attributes were tested using Welch's t-test, comparing the means of employees who left versus stayed. Categorical attributes were tested using the chi-square test of independence. A significance threshold of $p < 0.05$ was used to decide whether an attribute was statistically related to Attrition.

Attributes with no significant relationship to Attrition, such as Gender, Education, PerformanceRating, and NumCompaniesWorked, were dropped. Correlation coefficients were also reviewed alongside p-values to assess the strength, not just the significance, of each relationship.

3.4 Handling Class Imbalance

The target variable, Attrition, was imbalanced (approximately 84% stayed, 16% left). This class imbalance was distinguished from skewed distributions within individual features, which was treated separately and did not require correction. Each model addressed class imbalance using its own built-in mechanism: SMOTE (oversampling on the training set only) for Random Forest, scale_pos_weight for XGBoost, and auto_class_weights for CatBoost. A uniform imbalance-handling method was not applied across all three models, which is a limitation of the comparison.

3.5 Model Development

Three tree-based models were built: Random Forest, XGBoost, and CatBoost. These were selected for their ability to handle non-linear relationships and their reduced sensitivity to feature scaling and multicollinearity compared to linear models. Random Forest and XGBoost required categorical attributes to be one-hot encoded, while CatBoost used its native categorical handling and did not require manual encoding. Data was split into training and test sets (80/20) using stratified sampling to preserve the class distribution of Attrition in both sets.

4. Results & Evaluation

Model performance was compared using Precision, Recall, F1-score, and ROC-AUC, with particular attention to Recall and F1 on the Attrition class, since this reflects the model's ability to correctly identify employees who actually left.

Model	Class 1 Recall	Class 1 F1	ROC-AUC
Random Forest (SMOTE)	0.26	0.34	0.754
XGBoost	0.26	0.32	0.750
CatBoost	0.49	0.49	0.792

CatBoost outperformed both Random Forest and XGBoost across every metric, nearly doubling Recall and F1 on the Attrition class while also achieving the highest ROC-AUC. This is likely due to CatBoost's native categorical encoding, which preserves more signal from features like JobRole and Department compared to the one-hot encoding used for Random Forest and XGBoost. CatBoost was selected as the final model based on these results.

5. Limitations

- The dataset contains a relatively small number of employees who left (approximately 237 out of 1,470), limiting how much the models could learn about attrition patterns.
- The dataset is a single point-in-time snapshot, with no historical or trend data (e.g., changes in satisfaction or workload over time), which likely limits predictive power.
- Class imbalance was handled using a different method for each model (SMOTE for Random Forest, `scale_pos_weight` for XGBoost, `auto_class_weights` for CatBoost) rather than a single consistent method, which limits the fairness of the model comparison.
- Even the best-performing model (CatBoost) still misses roughly half of employees who actually left.

6. Conclusion

This project set out to build a machine learning model capable of predicting employee attrition and enabling early identification of at-risk employees. Statistical testing was used to identify which employee attributes were meaningfully related to attrition, and three tree-based models were trained and compared. CatBoost was selected as the final model, achieving the strongest performance across Recall, F1, and ROC-AUC, likely benefiting from its native handling of categorical attributes such as JobRole and Department. While the model demonstrates real predictive signal, its current recall on employees who leave would need further improvement before it could support production-level retention decisions.